

In [1]: `## importing libraries`

```
!pip install qiskit qiskit-aer matplotlib numpy
```

Requirement already satisfied: qiskit in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.3.1)
 Requirement already satisfied: qiskit-aer in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (0.17.2)
 Requirement already satisfied: matplotlib in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (3.10.8)
 Requirement already satisfied: numpy in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.4.4)
 Requirement already satisfied: rustworkx>=0.15.0 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from qiskit) (0.17.1)
 Requirement already satisfied: scipy>=1.5 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from qiskit) (1.17.1)
 Requirement already satisfied: dill>=0.3 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from qiskit) (0.4.1)
 Requirement already satisfied: stevedore>=3.0.0 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from qiskit) (5.7.0)
 Requirement already satisfied: typing-extensions in C:\Users\UNNATI NEGI\AppData\Roaming\Python\Python311\site-packages (from qiskit) (4.15.0)
 Requirement already satisfied: psutil>=5 in C:\Users\UNNATI NEGI\AppData\Roaming\Python\Python311\site-packages (from qiskit-aer) (7.2.2)
 Requirement already satisfied: python-dateutil>=2.8.0 in C:\Users\UNNATI NEGI\AppData\Roaming\Python\Python311\site-packages (from qiskit-aer) (2.9.0.post0)
 Requirement already satisfied: contourpy>=1.0.1 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.3.3)
 Requirement already satisfied: cyclor>=0.10 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (4.62.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.5.0)
 Requirement already satisfied: packaging>=20.0 in C:\Users\UNNATI NEGI\AppData\Roaming\Python\Python311\site-packages (from matplotlib) (26.0)
 Requirement already satisfied: pillow>=8 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (12.2.0)
 Requirement already satisfied: pyparsing>=3 in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (3.3.2)
 Requirement already satisfied: six>=1.5 in C:\Users\UNNATI NEGI\AppData\Roaming\Python\Python311\site-packages (from python-dateutil>=2.8.0->qiskit-aer) (1.17.0)

In [2]: `import numpy as np`

```
import matplotlib.pyplot as plt
```

```
from qiskit import QuantumCircuit
```

```
from qiskit_aer import AerSimulator
```

```
from qiskit.visualization import plot_histogram
```

In [3]: `!pip install pylatexenc`

Requirement already satisfied: pylatexenc in C:\Users\UNNATI NEGI\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.10)

```
In [4]: ## base circuit
def base_circuit():

    qc = QuantumCircuit(1,1)

    qc.h(0)
    qc.measure(0,0)

    return qc
```

```
In [5]: ##Circuit with X gate
def circuit_x():

    qc = QuantumCircuit(1,1)

    qc.h(0)
    qc.x(0)
    qc.measure(0,0)

    return qc
```

```
In [6]: ## circuit with Z gate
def circuit_z():

    qc = QuantumCircuit(1,1)

    qc.h(0)
    qc.z(0)
    qc.measure(0,0)

    return qc
```

```
In [7]: ## circuit with phase gate
def circuit_phase():

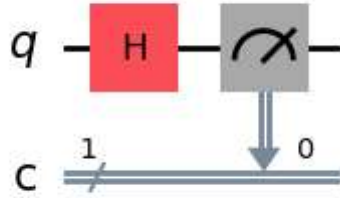
    qc = QuantumCircuit(1,1)

    qc.h(0)
    qc.p(np.pi/4,0)
    qc.measure(0,0)

    return qc
```

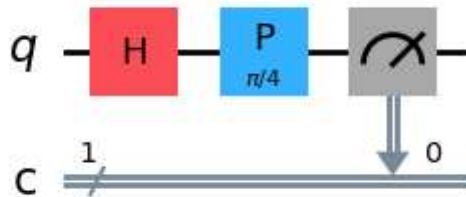
```
In [8]: base_circuit().draw('mpl')
```

Out[8]:



```
In [9]: circuit_x().draw('mpl')
circuit_z().draw('mpl')
circuit_phase().draw('mpl')
```

Out[9]:



```
In [10]: ## circuit simulator
simulator = AerSimulator()
```

```
In [11]: def run_circuit(qc):

    job = simulator.run(qc,shots=1000)
    result = job.result()

    counts = result.get_counts()

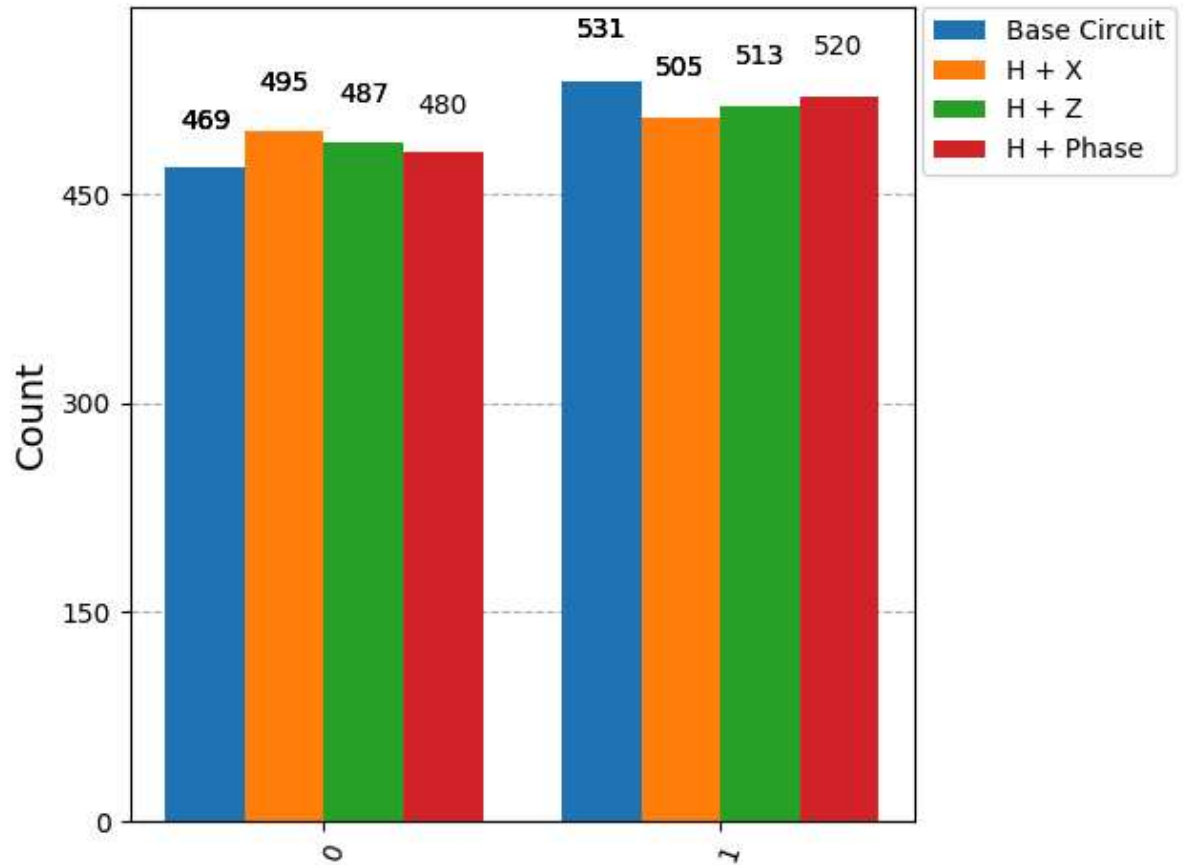
    return counts
```

```
In [12]: ## running circuits
counts_base = run_circuit(base_circuit())
counts_x = run_circuit(circuit_x())
counts_z = run_circuit(circuit_z())
counts_phase = run_circuit(circuit_phase())
```

```
In [13]: ## results
plot_histogram([
    counts_base,
    counts_x,
    counts_z,
    counts_phase
],
    legend=[
        "Base Circuit",
        "H + X",
        "H + Z",
```

```
"H + Phase"
])
```

Out[13]:



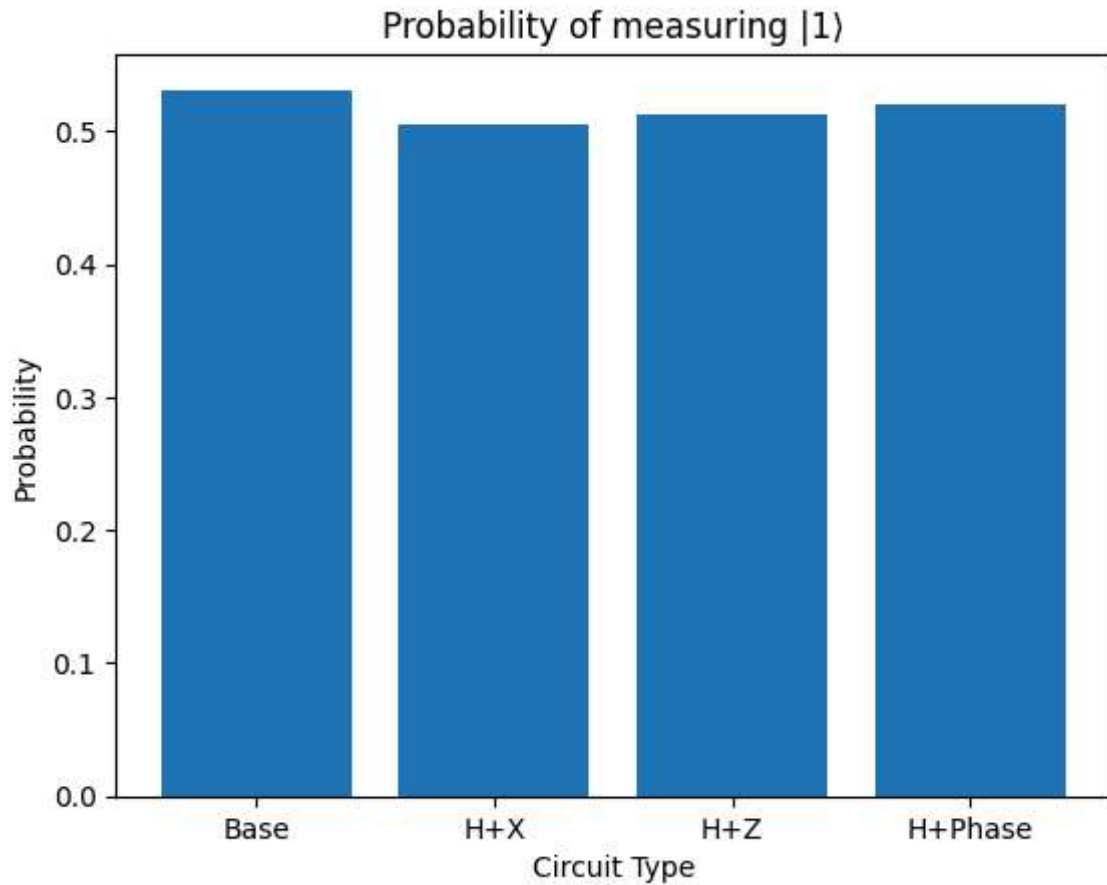
```
In [14]: ## Probability Comparison chart
circuits = ["Base", "H+X", "H+Z", "H+Phase"]

prob_1 = [
    counts_base.get('1',0)/1000,
    counts_x.get('1',0)/1000,
    counts_z.get('1',0)/1000,
    counts_phase.get('1',0)/1000
]

plt.bar(circuits,prob_1)

plt.title("Probability of measuring |1>")
plt.xlabel("Circuit Type")
plt.ylabel("Probability")

plt.show()
```



```
In [15]: ## sensitivity analysis
base_prob = counts_base.get('1',0)/1000

sensitivity = [
    abs(base_prob - prob_1[0]),
    abs(base_prob - prob_1[1]),
    abs(base_prob - prob_1[2]),
    abs(base_prob - prob_1[3])
]

plt.bar(circuits,sensitivity)

plt.title("Circuit Sensitivity Score")
plt.xlabel("Circuit Type")
plt.ylabel("Sensitivity")

plt.show()
```

